12.5.3 DATA CONTROL LANGUAGE

A **Data Control Language (DCL)** is a programming language used to control the access of data stored in a database. It is used for controlling privileges in the database (Authorization). The privileges are required for performing all the database operations such as creating sequences, views of tables etc.

SQL commands which come under Data Control Language are:

Grant	Grants permission to one or more users to perform specific tasks.
Revoke	Withdraws the access permission given by the GRANT statement.

12.5.4 TRANSACTIONAL CONTROL LANGUAGE

Transactional control language (TCL) commands are used to manage transactions in the database. These are used to manage the changes made to the data in a table by DML statements.

SQL command which come under Transfer Control Language are:

Commit	Saves any transaction into the database permanently.
Roll back	Restores the database to last commit state.
Save point	Temporarily save a transaction so that you can rollback.

12.5.5 DATA QUERY LANGUAGE

The Data Query Language consist of commands used to query or retrieve data from a database. One such SQL command in Data Query Language is

Select	It displays the records from the table.

12.6 Data Types

The data in a database is stored based on the kind of value stored in it. This is identified as the data type of the data or by assigning each field a data type. All the values in a given field must be of same type.

The ANSI SQL standard recognizes only Text and Number data type, while some commercial programs use other datatypes like Date and Time etc. The ANSI data types are listed in the following Table 12.1

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